

Hemanth S Vas

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Professional Summary

AI/ML Engineer specializing in production MLOps systems, agentic AI architectures, and end-to-end pipeline automation. Proven track record: solo-built MLOps infrastructure (0 to 90% production readiness), 99.9% uptime across 5+ microservices, 50K USD client retention, and 40% performance gains. Expert in AST/CST-based code analysis, distributed systems, and cross-functional leadership.

Work Experience

AI Engineer Intern

August 2024 to Present

PromptIQ Technologies Pvt. Ltd.

Bengaluru, India

- Architected and solo-deployed production MLOps automation pipeline from 0 to 90% readiness, designed 8-stage architecture, implemented 62 automated tests with 100% pass rate, built CI/CD with GitLab, integrated Celery and Redis job queue, and deployed structured JSON logging achieving 99.9% pipeline reliability serving 500+ daily fine-tuning jobs
- Designed and built agentic coding system for codebase analysis, architected AST/CST parsing pipeline using tree-sitter, engineered vector embedding strategy with ChromaDB for 10K+ code chunks achieving sub-50ms semantic search, implemented LangGraph-based multi-agent orchestration, and deployed async Celery workflows processing 2TB+ codebases with 95% accuracy
- Drove RAG pipeline product strategy serving legal and financial clients, architected hybrid RAG and fine-tuning approach, and launched release achieving 92% accuracy with 500+ daily queries and 50K USD client retention
- Architected OAuth2 authentication roadmap, shipped Google and Azure AD integration securing 5000+ users with SOC 2 and GDPR compliance and sub-200ms latency

AI Engineering Intern

August 2023 to January 2024

Simulation AI Internships

Remote

- Oversaw end-to-end ML model deployment across 4 production environments, shipped Docker-based microservices reducing deployment failures by 15% and rollback time by 40%
- Designed and implemented Airflow ETL pipeline for 2TB+ data processing, improved throughput by 15% and reduced data quality errors by 25%

Key Projects

MLOps LLM Automation Pipeline | *Python, FastAPI, Celery, Redis, ChromaDB, pytest, GitLab CI/CD*

- Solo-architected production MLOps system, designed 8-stage pipeline including ingestion, preprocessing, fine-tuning, RAG, evaluation, and monitoring, implemented 62 tests with 100% pass rate, and deployed CI/CD automation achieving 99.9% reliability
- Engineered production infrastructure including Celery and Redis distributed job queue, implemented structured JSON logging, and documented performance benchmarks with RAG p50 at 25-50ms and p95 at 80-150ms

Agentic Code Analysis System | *Python, tree-sitter, LangGraph, ChromaDB, Celery, FastAPI*

- Architected end-to-end AST/CST-based code intelligence pipeline, designed multi-language parser using tree-sitter, implemented ChromaDB vector store with 10K+ indexed code segments achieving sub-50ms p95 retrieval latency
- Built multi-agent orchestration with LangGraph, designed agent roles including code reviewer, refactorer, and tester, implemented async Celery workflows processing 2TB+ repositories, achieving 95% accuracy in architectural recommendations

Technical Skills

- Languages:** Python, Go, Java, SQL, JavaScript, TypeScript
- ML/AI Frameworks:** TensorFlow, PyTorch, Keras, scikit-learn, XGBoost, HuggingFace Transformers
- LLM and NLP:** LangChain, LangGraph, LlamaIndex, OpenAI API, RAG, FAISS, Pinecone, ChromaDB, Semantic Search
- Agentic AI:** Multi-agent orchestration, ReAct pattern, LangGraph workflows, CrewAI, AutoGen
- Code Analysis:** AST/CST parsing using tree-sitter and ast module, static analysis, code embeddings
- Cloud and Infrastructure:** GCP, AWS, Azure, Docker, Kubernetes
- Data Engineering:** MongoDB, PostgreSQL, Redis, Elasticsearch, Kafka, Airflow, FastAPI
- MLOps and Testing:** MLflow, Weights and Biases, Prometheus, Grafana, pytest, CI/CD using GitLab, Git, Celery

Education

Visvesvaraya Technological University

Hassan, Karnataka, India

Bachelor of Engineering in Artificial Intelligence and Machine Learning; CGPA: 7.8/10.0

November 2022 to May 2026